IE 5311 Principles of Optimization — exam sheet

Slide numbers in brackets. Grading [6]: 40% problem to formulation to solution, 60% understand the method. Theory is the bigger half.

1. CLASSIFY FIRST [8-10]

Applied to diet [18], shortest path [27], DFJ [32]. Expect it on the exam. Nine axes:

```
1 constrained / unconstrained
2 variables: continuous | discrete | mixed
3 variable size: compact | huge finite | infinite-dim
4 obj & constr type: linear | convex | nonconvex
5 constraint size: compact | huge finite | semi-infinite
6 objective size: single | multi
7 parameters: deterministic | stochastic
8 stages: one | multi
9 players: single | two | multi, simultaneous | sequential
```

2. WRITE EVERY MODEL IN THIS ORDER

Sets (with index) → Parameters (with units) → Variables (domain + units) → Objective as a sum → Constraints, each with a *for all* → Domains stated separately. Never enumerate numeric rows. **Define a derived set by its formula, never by a name.**

3. FIVE NOTIONS OF OPTIMAL, ONE NEWSVENDOR [39-59]

```
expectation min E[f(x,xi)] SAA over scenarios
robust min max_{xi in Xi} f worst case [41,42]
CVaR min CVaR_beta(f) tail average [58]
chance min c'x, P(feas) >= gam [59]
DRO min max_{P in Pset} E_P change the measure [53]
```

Same word problem, five meanings of optimal. Closed form for the single-item newsvendor: q* = F^-1(b / (b+h)), b underage, h overage. Wait-and-see (decide after xi) ≤ here-and-now.

4. RISK MEASURES [55-57]

```
VaR_b(Z) = min{ t : P(Z <= t) >= b }
CVaR_b(Z) = min over eta of eta + (1/(1-b)) E[(Z-eta)+]
```

CVaR is an **infimum**, so to impose CVaR ≤ Lambda you may let eta be a free decision variable: no inner minimisation needed, and the reformulation is exact.

```
eta + (1/(1-b)) sum_s p_s u_s <= Lambda
u_s >= Z_s - eta, u_s >= 0, eta free
```

Dual / risk-envelope form, CVaR as a robust problem:

```
CVaR_b(Z) = max_{q in Q} sum_j q_j Z_j
Q_b = { q >= 0 : sum_j q_j = 1, q_j <= p_j/(1-b) }
```

So constraint generation for CVaR runs over **Q**, not over scenarios: every scenario stays in the master, so the master stays bounded. Always VaR_b ≤ CVaR_b and E[Z] ≤ CVaR_b.

5. PROBABILITY [43-57]

Space (Omega, F, P). F is a sigma-algebra: contains Omega, closed under complement and countable union. Pushforward of X: Omega → R is P_X(B) = P(X^-1(B)). Expectation as **dual pairing** <P, Z> [52] is the hinge: expectation, LP duality and conic duality are one idea in three settings.

6. INDICATOR AND BIG-M

Over a box l ≤ x ≤ u the extrema are separable:

```
Vmax(a) = sum_i max(a_i l_i, a_i u_i) at x_i = u_i if a_i>=0
Vmin(a) = sum_i min(a_i l_i, a_i u_i) at the other corner
z=1 => a'x<=b : a'x <= b + M(1-z), M = max{0, Vmax(a)-b}
z=0 => a'x>=b : a'x >= b - N z, N = max{0, b-Vmin(a)}
z_k=z_l=1 => c'x<=d : c'x <= d + M(2 - z_k - z_l)
violate only if active : g'x <= h + M sum_k z_k
```

The pair and cover forms need **no extra binary**. Those M are smallest, and tight: the witness is the box corner above.

A two-way equivalence needs a tolerance: add g'x ≥ h + eps – N(1–z_k) for all k. **Why eps cannot go:** {g'x > h} is open, while a MILP feasible set is a finite union of closed polyhedra, hence closed. No exact MILP exists.

7. CHANCE CONSTRAINTS

Joint, finite support, one binary per scenario:

```
u_js <= (1-th) d_js + M_js (1 - z_s) for all j, s
sum_s p_s z_s >= gamma
smallest M_js = th * d_js when 0 <= u_js <= d_js
```

M depends on **both** j and s; one global M is valid but loose. Joint ⇒ individual by monotonicity. The converse **fails**: the union bound gives only P(cap A_j) ≥ 1 – |J|(1–gamma), and that is tight, so build the counterexample with disjoint failure events. Bonferroni fix: impose each region at level 1 – (1–gamma)/|J|.

8. ROBUST AND CONSTRAINT GENERATION

A linear max over a polytope attains at an **extreme point**, so replace the inner max by one row per vertex. Too many vertices → master + separation subproblem [33,34], with finite convergence and correctness proofs. **Seed the master with a point inside Xi**: seeding outside makes the master a restriction, and it converges *above* the true optimum.

9. EXACTNESS AND INTEGRALITY [25,26]

A is totally unimodular if every square submatrix has determinant in {0, +1, –1}. TU with integer b makes every vertex of {Ax ≤ b, x ≥ 0} integral, so the LP relaxation is exact — which is why shortest path needs no binaries. Node-arc incidence matrices of digraphs are TU. DFJ vs MTZ [30]: same feasible solutions, different strength; DFJ has exponentially many subtour rows.

10. TWO-STAGE STOCHASTIC

```
min c'x + sum_s p_s Q(x, xi_s), y_s chosen AFTER xi_s
```

Scenario-index every second-stage variable. A bound that must hold in every realisation is a *robust* constraint and does not belong in an average-case model. For feasibility under uncertainty use a chance constraint or recourse.

11. LINEAR ALGEBRA GATE [7]

Basis = independent + spanning; dim = size of a basis. The vector is the object, the coordinate list [v]_B is not. Independence, two equivalent definitions: (i) no v_i lies in the span of the others; (ii) sum a_i v_i = 0 forces every a_i = 0. Prove the equivalence by **contraposition** both ways; invertibility of a nonzero scalar is the only thing used, so it needs a field. Change of basis: the columns of P are the new basis in old coordinates; B' is a basis iff det P ≠ 0; solve P a = [v]_B. T linear iff T(lf + mg) = l T(f) + m T(g); the columns of its matrix are the images of the basis. ker T = {v : Tv = 0}, range T = span of those images, and **dim ker + dim range = dim V**.

12. GROUPS

Closure, associativity, identity, inverse. Commutativity is **not** required. Inverse is unique: b = be = b(ac) = (ba)c = ec = c, hence (a^-1)^-1 = a. GL_n(R) is a group since det(AB) = det A det B ≠ 0, non-abelian for n ≥ 2. The rotations of a square are Z/4Z, cyclic, generated by the quarter turn.

13. WHAT HAS COST MARKS

Naming a derived set instead of writing its formula. A robust bound inside a stochastic model. Generating CVaR cuts over scenarios rather than over the risk envelope. Hedging a performance claim is fine when the hedge is true.

14. THE PART 1 PROBLEM LIST [12-61]

Outline [12]: Part 1 classic problems, Part 2 linear programming, Part 3 conic programming. Topics marked ** are optional: write a short review for 1 point each, cap 10.

```
diet [14-19] continuous, LP, determ., 1-stage, 1-player
shortest path [20-27] binary compact, linear, determ.
TSP / DFJ [28-35] huge but finite constraint set
facility loc [36-38] two-player
newsvendor [39-59] five criteria, see box 3
stoch. shortest path [60]
```

Stigler pruned 77 foods to 15 by hand and got $39.93; Dantzig solved it exactly at $39.69 by simplex seven years later. The 24 cents started the field. Convex preview [61]: portfolio (convex, multi-objective), inventory (stochastic expectation), robust shortest path (worst case, semi-infinite), multi-period inventory (MDP), bilevel pricing (sequential two-player).

15. GRAPH VOCABULARY [20-24]

Undirected: neighbourhood, degree, cut, walk, trail, path, cycle. A walk repeats anything, a trail repeats no edge, a path repeats no vertex. Directed: in- and out-neighbourhood, in- and out-degree, in-cut and out-cut. Flow balance is written on the in-cut minus the out-cut at each node.

16. PROOF MOVES THAT KEEP COMING BACK

Contraposition for an equivalence of two definitions: show each fails exactly when the other does. **Uniqueness** by writing one object two ways and collapsing the chain, as with the group inverse. **Monotonicity** of probability for one direction of joint versus individual. **Extreme point** arguments to turn an infinite max into finitely many rows. **Closedness** to prove no exact MILP exists. State what each step uses; the marks are in the reason, not the result.